

# ML Specification Report — playground-series-s6e7

## Student Health-Risk Prediction (3-class) • AIDE agent (20-step tree search, Claude Sonnet via shim)

Scores grounded exclusively in `facts_aide.json` for run `2-rational-cryptic-manatee`; pipeline configuration read from AIDE's own `best_solution.py` and `report.md`, step-level intent from `journal.json`, and task metadata from `competitions/playground-series-s6e7/config.yaml`.

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## Overview

The task is to assign each of 295,753 test subjects to one of three health classes — at-risk / unhealthy / fit — from lifestyle and physiological features, scored by **balanced accuracy**. This run was executed by **AIDE**, a tree-search coding agent: each step writes one complete `solution.py`, runs it in a sandbox under a 30-minute wall clock, reads back the printed metric or the traceback, and mutates. Over **20 steps** it produced **17 scored solutions and 3 crashes**, and its champion — found at **step 17** — is a multinomial **logistic regression stacker** over logit-transformed base-model probabilities, scoring **CV balanced accuracy 0.94981** locally and **0.94982** on the public leaderboard, for **rank 895 / 2542** (64.8th percentile).

**A structural caveat governs this entire report and must be stated before anything else.** The input directory handed to AIDE for this competition did not contain only `train.csv`, `test.csv` and `sample_submission.csv`. It also contained **seven pre-computed out-of-fold / test probability archives** — `oof_cat.npz`, `oof_et.npz`, `oof_lgbm.npz`, `oof_lgbm_seed2024.npz`, `oof_lgbm_tuned.npz`, `oof_mlp.npz`, `oof_xgb.npz` — i.e. the OOF predictions of a seven-model base ensemble that AIDE did not train. AIDE noticed them at step 0 and, entirely reasonably, built a stacker on top of them; it **never trained a single base model from raw features in the whole run**. Everything below therefore describes a meta-

learning run, not an end-to-end modelling run, and the comparison against the other lanes for this competition is not a like-for-like agent comparison. This is a lane-isolation failure in the data staging, and it is reported here rather than papered over.

**Why it matters.** Screening a large population into health-risk tiers from cheap lifestyle signals is a triage problem where the minority classes are the ones that matter — which is exactly why the metric is balanced accuracy and not accuracy. Per `config.yaml` the class prior is **at-risk 0.859 / unhealthy 0.084 / fit 0.058**, so a majority-class predictor scores the 0.333 floor; every point above that comes from correctly surfacing the two small classes.

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## Data

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**Purpose of Data.** Classify `health_condition` into three levels. **Data Format** is **tabular CSV** for the raw features plus **seven NumPy .npz archives** of pre-computed probabilities. **Data Volume**, per `config.yaml`, is **690,088 training rows / 295,753 test rows**; the raw feature set is 7 numeric columns (`sleep_duration`, `heart_rate`, `bmi`, `calorie_expenditure`, `step_count`, `exercise_duration`, `water_intake`) and 6 categorical columns (`diet_type`, `stress_level`, `sleep_quality`, `physical_activity_level`, `smoking_alcohol`, `gender`).

**Data Quality** is, in this lane, almost entirely irrelevant — and that is itself the finding. Because AIDE worked on the stacked probability matrix, it never inspected missingness, distributions, or train↔test shift in the raw columns; there is no EDA artifact of any kind, since the agent's loop has no EDA stage. The one piece of data-quality work it actually did was a schema-discovery bug hunt: at step 6 the script crashed with a `ValueError` because AIDE's key-matching heuristic looked for `.npz` keys containing substrings like "oof" or "prob", while the archives in fact use the bare keys "oof" and "test". Step 8 diagnosed and fixed this by matching on array shape (`shape[0] == n_train / == n_test`) rather than key name — and that shape-based resolver is what survives into the champion script.

The defining wrinkle of the task, **severe three-class imbalance**, was handled correctly and early: `class_weight="balanced"` at step 2 lifted the score from 0.92475 to 0.94971, by far the largest single gain of the run. AIDE also tested the obvious alternative — post-hoc prior correction, dividing predicted probabilities by empirical class frequencies — at step 4, and it collapsed the score to 0.8973, because it double-corrects an imbalance that `class_weight` has already removed. That is a genuine controlled ablation, and it is one of only two in the run.

**Annotation Guidelines.** The label is `health_condition`  $\in$  {at-risk, unhealthy, fit}, integer-encoded with `LabelEncoder` for modelling and inverse-transformed back to the original strings for submission. The submission is a **hard class label** per row, not a probability — balanced accuracy is computed on the argmax decision.

**Feature Set.** The champion's feature matrix is not the competition's features at all:

| Group                    | Content                                                                      | Width |
|--------------------------|------------------------------------------------------------------------------|-------|
| Base-model probabilities | 7 archives $\times$ 3 class probabilities, resolved by row-count shape match | 21    |
| Transform                | $\text{logit}(p) = \log(p / (1 - p))$ with $p$ clipped to $[1e-6, 1 - 1e-6]$ | 21    |
| Scaling                  | <code>StandardScaler</code> fitted on the logit matrix                       | 21    |
| Raw lifestyle features   | <b>none</b> — tested at step 18, rejected                                    | 0     |
| Consensus statistics     | <b>none</b> — tested at step 12, rejected                                    | 0     |

The **logit transform is the single most important modelling decision in the run**: raw probabilities are bounded and pile up near 0 and 1, whereas their log-odds are unbounded and roughly linear in the quantity a logistic meta-learner is actually fitting. Step 2 (balanced weights, raw probabilities) scored 0.94971; step 3 (balanced weights, logit features) scored 0.94979. Every subsequent enrichment was rejected on the same folds: appending the raw lifestyle features scored 0.94973 (step 18), appending per-class mean/std/min/max and per-model entropy scored 0.94967 (step 12), and concatenating raw and logit representations into 42 columns scored 0.94972 (step 19). All three lost to the plain 21-column logit stack.

**Splitting strategy.** `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)` for the reported metric, with an inner **3-fold** stratified search over the regularization strength `C` to keep tuning inside the 30-minute box. This is the run's clearest methodological weakness and AIDE's own report names it: the `C` chosen by the inner search and the score reported by the outer 5-fold evaluation come from the same training data, so the reported 0.94981 is mildly optimistic as a generalization estimate. The competition was still open at submission time (deadline 2026-07-31 per `config.yaml`), so `facts_aide.json` carries a **public LB score but no private LB score**.

## Models & Architecture

**Purpose of Architecture.** Combine seven frozen base-model probability vectors into a single three-class decision that maximizes balanced accuracy. **Architecture Type** is a **multinomial (softmax) logistic regression meta-learner** — a linear model, not an ensemble of trees, and not a neural network. AIDE reached this by elimination rather than assumption: it tried LightGBM as meta-learner (0.87968), an `MLPClassifier` with 32 hidden units (0.87862), LDA (0.91463), QDA (0.94146), a LogReg+LDA 50/50 blend (0.92467), an alpha-tuned LogReg+QDA blend (0.94979) and `LinearSVC` with hinge loss (0.94471). **Every non-linear and every alternative-linear meta-learner lost to plain logistic regression**, and the tuned blend search drove its own mixing weight to the pure-LogReg endpoint.

**Input Format** is a dense float64 matrix of standardized log-odds; **Input Dimension is 21** (7 models  $\times$  3 classes). **Architecture Description:** `LogisticRegression(max_iter=1000, class_weight="balanced", C=best_C, solver="lbfgs", n_jobs=-1)`, where `best_C` is selected from the grid `{0.01, 0.03, 0.1, 0.3, 1.0}` by the inner 3-fold search (AIDE's report records 0.01 as the winner — the most heavily regularized point

on the grid, which is consistent with 21 near-collinear features on 690k rows). **Model Complexity** is trivially small:

```
3 classes × 21 features + 3 intercepts = 66 parameters
```

Sixty-six parameters sitting on top of seven frozen base models. The champion node is step 17, and it carries no blend weights of its own — the "ensembling" all happened upstream in base models AIDE did not build.

## Training procedures

### The search trajectory

Of the **20 steps** in the budget, **17 produced a score and 3 crashed**:

```
20 - 17 = 3 buggy steps
```

The failures split into one logic error and two budget overruns ( `buggy_exc_types: {"ValueError": 1, "TimeoutError": 2}` ), at the three steps absent from the metric trajectory — **steps 6, 10 and 14**. Step 6 is the `ValueError` (the `.npz` key-matching bug described above, fixed at step 8). Steps 10 and 14 are the timeouts, each hitting the 1800 s harness wall ( `exec_time_max_s = 1800.1 s` ), so together they burned

```
2 × 1800 = 3600 s
3600 / 6312.7 ≈ 0.5703 → 57.0% of all execution time
6312.7 / 3600 ≈ 1.75 h total
```

**More than half of this run's compute produced no score at all.** Both timeouts have the same root cause class — trying to make a linear model expensive. Step 10 expanded the 21 logit features into pairwise interactions with `PolynomialFeatures(degree=2, interaction_only=True)` , taking the design matrix to 231 columns on 690,088 rows and stalling L-BFGS; step 14 ran an exhaustive 25-combination elastic-net grid with the `saga` solver on the same data. AIDE diagnosed both correctly and repaired both — step 15 shrank the search space and switched to a single train/validation split for the coarse stage, and step 17 simply deleted the polynomial expansion.

And that is the run's most striking structural fact: **the champion is a bug-fix node**. Step 17's plan is not an improvement proposal, it is a post-mortem ("the root cause of the timeout is the `PolynomialFeatures` step ... the fix is to drop it"), and the stripped-down repair it produced — 21 logit features, standardized, L2, `lbfgs` , `C=0.01` — turned out to be the best solution AIDE found. The agent's best idea was the removal of its own worst idea.

| Step | What AIDE changed                                   | CV balanced accuracy |
|------|-----------------------------------------------------|----------------------|
| 0    | LogReg stacker on raw base probabilities (baseline) | 0.92475              |
| 1    | LightGBM meta-learner + raw lifestyle features      | 0.87968              |

| Step | What AIDE changed                                                                            | CV balanced accuracy                         |
|------|----------------------------------------------------------------------------------------------|----------------------------------------------|
| 2    | <code>class_weight="balanced"</code> + tuned <code>C</code> , raw probabilities              | 0.94971                                      |
| 3    | <b>logit transform of the 21 probability features</b>                                        | 0.94979                                      |
| 4    | post-hoc prior correction on the argmax                                                      | 0.8973                                       |
| 5    | LDA meta-learner                                                                             | 0.91463                                      |
| 6    | QDA meta-learner                                                                             | ValueError ( <code>.npz</code> key matching) |
| 7    | <code>MLPClassifier</code> , 32 hidden units                                                 | 0.87862                                      |
| 8    | bug fix: shape-based <code>.npz</code> resolution → QDA runs                                 | 0.94146                                      |
| 9    | LogReg + LDA 50/50 blend                                                                     | 0.92467                                      |
| 10   | pairwise interaction expansion (21 → 231 features)                                           | TimeoutError                                 |
| 11   | LogReg + QDA alpha-tuned blend                                                               | 0.94979                                      |
| 12   | + per-class consensus statistics and entropy                                                 | 0.94967                                      |
| 13   | L1 penalty via <code>saga</code>                                                             | 0.9498                                       |
| 14   | elastic-net, 25-combination <code>saga</code> grid                                           | TimeoutError                                 |
| 15   | timeout repair: shrunk search, holdout coarse stage                                          | 0.94974                                      |
| 16   | <code>LinearSVC</code> , hinge loss                                                          | 0.94471                                      |
| 17   | <b>timeout repair: drop interactions, scale, L2 <code>lbfgs</code> , <code>C=0.01</code></b> | <b>0.94981</b>                               |
| 18   | + raw lifestyle features appended to the logit stack                                         | 0.94973                                      |
| 19   | raw + logit dual representation (42 features)                                                | 0.94972                                      |

The trajectory spans a wide band because of the failed model-family excursions —

$$0.94981 - 0.87862 = 0.07119$$

— but the useful range is minute. The two decisions that mattered both landed by step 3, and their combined effect is

$$0.94981 - 0.92475 = 0.02506$$

After step 3 the search is flat to the fourth decimal: step 3 reached 0.94979, and the remaining **fourteen scored steps bought**

$$0.94981 - 0.94979 = 0.00002$$

That is an honest plateau. AIDE spent roughly three quarters of its budget confirming, across six distinct meta-learner families and four feature-enrichment variants, that a tuned linear model on 21 logit features could not be beaten — a negative result that is genuinely informative, but a very expensive way to obtain it. The agent had no mechanism to notice it had converged and stop.

## Training configuration

The **Loss Function** is multinomial cross-entropy (softmax log-loss) with **balanced class weights**, i.e. each class's contribution is reweighted by the inverse of its frequency — the single change that took the run from 0.92475 to 0.94971, and mandatory here because the metric is balanced accuracy on an 0.859 / 0.084 / 0.058 prior. The **Optimization Algorithm** is **L-BFGS**, a full-batch quasi-Newton method, with `max_iter=1000` for the final fits and `max_iter=500` during the inner `c` search; regularization is L2 with strength `1/c`. There is **no Learning Rate** in the usual sense (N/A — L-BFGS selects its own step length by line search; the tunable analogue is the regularization strength `c`, grid-searched over {0.01, 0.03, 0.1, 0.3, 1.0}), **no Learning Rate Scheduler** (N/A — convergence is governed by the L-BFGS tolerance and `max_iter`, not a schedule), and **no Batch Size** (N/A — L-BFGS consumes all 690,088 rows per iteration; the run is full-batch by construction).

**Training Duration** across the whole search was **6,312.7 s** of execution time over 20 nodes, mean **315.6 s**, max **1,800.1 s** (the harness cap). Per-node wall time for the champion is not carried in `facts_aide.json` — **not recorded**. **Training Memory** was never instrumented — **not recorded**; the working set is a  $690,088 \times 21$  float64 matrix plus the seven source archives.

**Transfer Learning** is the field where this competition is genuinely anomalous. Formally it is N/A (no pretrained weights), and AIDE has **no cross-competition memory** — every run starts cold, its only continuity being the within-run journal that lets each plan quote its incumbent's score ("the best result so far (0.94979)..."). But in substance, the seven pre-computed `.npz` archives are a transfer of learned representations from outside the run, and AIDE consumed them. It is the reason a 66-parameter linear model reaches the 64.8th percentile. **Data Augmentation** is N/A (tabular); the analogues AIDE tried — feature enrichment from the raw columns, cross-model consensus statistics, dual raw+logit encoding, pairwise interactions — were tested at steps 18, 12, 19 and 10 respectively and **all four were rejected or timed out**.

**Reproducibility Standards** are adequate for a deterministic model and no more. `random_state=42` is fixed on both the inner 3-fold and outer 5-fold splits; logistic regression under L-BFGS is deterministic given its inputs, so there is no seed-sensitivity to control. But there is **no rebuild or reproduction gate** — AIDE never retrains its champion to verify the cached score — and, decisively, **the run is not reproducible from the competition data alone**: it depends on seven `.npz` artifacts produced outside the run, whose provenance and generating code are not part of this lane. Re-running `best_solution.py` against a clean competition download would fail at the first `np.load`.

## Inference procedures

**Decision Threshold** is N/A in the binary sense — the submission requires a hard three-class label, and the rule used is a plain **argmax over the softmax outputs** of the final meta-model. AIDE explicitly tested the alternative: step 4's prior-corrected argmax (dividing by empirical class frequencies before taking the maximum) scored 0.8973 against the 0.94979 of the uncorrected rule, so the naive argmax was retained. No per-class threshold search was ever run; AIDE's own report lists it as future work.

**Post-processing** is `LabelEncoder.inverse_transform` back to the original class strings, then column reordering to match `sample_submission.csv`. Note that unlike the CV loop, the **final model is refit once on all 690,088 training rows** (not a fold ensemble) before predicting the test set — so the submitted predictions come from a single logistic regression, while the reported 0.94981 comes from five fold models. **Inference Duration** and **Inference Memory** were not profiled — **not recorded**; scoring 295,753 rows through a 66-parameter linear model is negligible.

## Evaluation & Benchmarking

**Task for Performance Evaluation.** Assign each of the 295,753 test subjects to one of three health classes, scored by balanced accuracy (the unweighted mean of per-class recall).

**Performance Metrics.** AIDE's champion scores **CV balanced accuracy 0.94981** locally and **0.94982** on the public leaderboard, for **rank 895 / 2542** = the **64.8th percentile**. The CV-to-LB agreement is as close as this metric can plausibly get:

$$0.94982 - 0.94981 = 0.00001$$

which says the 5-fold stratified estimate is essentially unbiased here — unsurprising with 690k training rows and 295k test rows, where both estimates are tightly concentrated. The competition was still open, so `facts_aide.json` carries **no private LB score** and no percentile shift can be checked.

The full arc of the run, from first scored solution to champion, is

$$0.94981 - 0.92475 = 0.02506$$

of which the balanced class weighting and the logit transform account for essentially all — both landed by step 3, and the following fourteen scored steps added 0.00002.

**Performance Benchmarking.** A numeric three-way table **cannot be given for this competition**. `facts_aide.json` 's score table records the verdict fields `local_winner: mine` and `lb_winner: mine (pub)` — i.e. the from-scratch agent won both the local and the public-LB comparison — but it does **not** carry `mine_priv` or `nvidia_priv` values for s6e7, and inventing comparator numbers is not permitted. What can be said is:

| Agent              | Approach                                             | Public LB balanced accuracy | Note                                           |
|--------------------|------------------------------------------------------|-----------------------------|------------------------------------------------|
| AIDE               | 20-step search → LogReg stacker on 21 logit features | 0.94982                     | rank 895 / 2542, 64.8th pct                    |
| From-scratch agent | not carried in <code>facts_aide.json</code>          | —                           | recorded as <code>lb_winner: mine (pub)</code> |
| NVIDIA             | not carried in <code>facts_aide.json</code>          | —                           | no figure available                            |

Two honest qualifications close this out. First, the comparison is **not like-for-like**: AIDE was handed a seven-model OOF ensemble in its input directory and spent all twenty steps stacking it, so what is being compared is one agent's end-to-end pipeline against another agent's meta-learner over borrowed base models. Second, AIDE's own contribution to the score is small and quickly exhausted — two good decisions by step 3, then a fourteen-step plateau worth 0.00002, with 57.0% of its compute lost to two self-inflicted timeouts on a problem whose winning model has 66 parameters.